

T2DyVec: Leveraging Sparse Images and Controllable Text for Dynamic SVG

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ABSTRACT

In this work, we introduce a controlled dynamic vector graphic generation method. While existing work mostly focuses on text-based generation of single-frame images, dynamic images, or single-frame vectors, there is a lack of research on generating dynamic vectors with complex elements and diverse styles. This is due to the unique challenges posed by dynamic vectors, which require coherent and seamless transitions of vector parameters between frames. To address these challenges, we propose T2DyVec, which leverages text prompts and sparse images as a control for vector generation. It incorporates Vector Consistency, Semantic Tracking, and VPSD to optimize the diffusion model for vector parameters, enabling the generation of multi-frame dynamic coherent vectors. This approach can significantly optimize creative workers' workflow, facilitating generation and further editing.

CCS CONCEPTS

• **Computing methodologies** → *Image representations; Shape inference; Interest point and salient region detections; Procedural animation.*

KEYWORDS

Dynamic Scene, Text to Vector Graphics, Controllable Generation, SVG

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1 INTRODUCTION

It's undeniable that painting has long been a vital medium for expressing and conveying information. With the advent of the digital age, many paintings have transitioned to electronic formats, with Scalable Vector Graphics (SVG) playing a crucial role in electronic painting. SVG defines 2D graphics and supports interaction and

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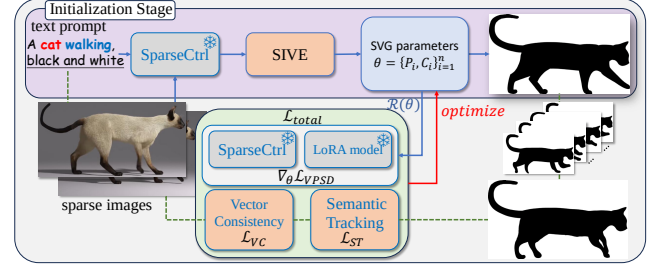


Figure 1: The pipeline of T2DyVec. The method takes a text description and several images as controllable conditions and produces coherent dynamic vector graphics.

animation, offering a creative platform and enhancing the editing feasibility of creations for artists and designers alike. The objective of this work is to investigate the generation of dynamic vectors using a diffusion model, with the guidance of sparse images and text inputs, while retaining the benefits of vectors.

The above-mentioned task is challenging in two aspects: (i) dynamic SVGs should be coherent as much as possible at the geometric level among different frames; (ii) dynamic SVGs should be semantically consistent with sparse image inputs and text guidance. In light of the foregoing challenges, we introduce T2DyVec to produce controllable, coherent, and semantically aligned dynamic vectors.

In particular, we introduce an as-rigid-as-possible consistency objective for each vector. This objective encourages that every single vector does not deviate too much from proceeding ones during the optimization process, therefore guaranteeing the overall rigidity of a vector. For semantic alignment, we use interpolation to track sparse images and then match vectors so that the semantic information of the image sequence is accurately captured by the temporal vector sequences. Furthermore, we also leverage VPSD to enhance our overall approach and employ LoRA to help manage the optimization time more effectively.

2 METHODOLOGY

Given a sequence of S sparse images $\{I_t\}_{t=1}^S$ and a text prompt y , T2DyVec generate the corresponding sequence of T dynamic vectors with n paths $\{\theta_t\}_{t=1}^T = \{\{P_{t,i}, C_{t,i}\}_{i=1}^n\}_{t=1}^T$, where $S \ll T$. Each dynamic vector consists of m control points $P_{t,i} = \{p_j\}_{j=1}^m = \{(x_j, y_j)\}_{j=1}^m$ with shared color contributes $C_{t,i} = \{r, g, b, a\}$.

Firstly, we introduce Vector Consistency, which is designed to ensure rigid consistency across each vector path, and therefore the coherence of the dynamic vectors. According to the theory

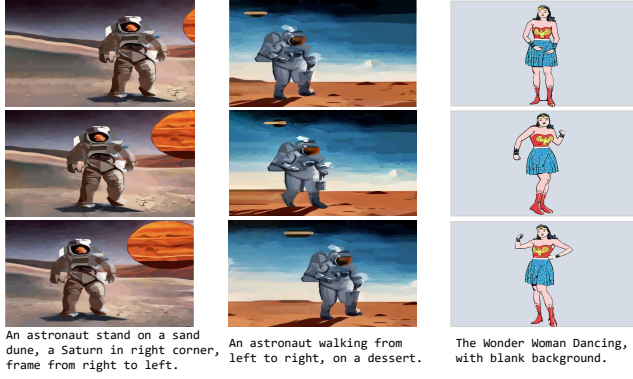


Figure 2: Results with various styles.

of ARAP [Sorkine and Alexa 2007], we establish the rigid consistency of the vector path by considering any three parameter points as the vertices of a triangle. Since the vector path undergoes rigid changes from frame to frame, we simplify the problem to ensure that the side lengths of the triangles formed by corresponding points in two neighboring frames exhibit similar properties. Thus, for the i -th path at frame t , we have $\mathcal{L}_{VC} = \sum_i \sum_{(\alpha, \beta, \gamma)} d(p_\alpha, p_\beta) + d(p_\alpha, p_\gamma) + d(p_\beta, p_\gamma)$, where $d(p_m, p_n) = \text{abs}(\|p_{t,m} - p_{t,n}\|_2 - \|p_{t-1,m} - p_{t-1,n}\|_2)$, and $(p_\alpha, p_\beta, p_\gamma)$ represents any three distinct parameter points selected from the i -th vector path. This ensures that each triangle remains as rigidly consistent as possible during optimization. When all triangles on a path satisfy this rigid consistency, we assume that the entire path similarly satisfies this property. This approach ensures the coherence of the vector sequence.

Subsequently, we introduce Semantic Tracking to maintain the semantic consistency of vectors. In essence, we sample a subset of the feature points $p_{t,j} \in \{P_{t,i}, \psi(P_{t,i})\}$, where $P_{t,i}$ represents the feature points on the i -th vector path and the $\psi(P_{t,i})$ represents the feature points in the plane of the i -th vector path at the t -th frame. We then employ a tracking algorithm $T(\cdot)$ to imbue it with semantic relevance. However, since the input is a sequence of sparse images, there is not necessarily an image at the current t -th frame for a conditionally supervised generation, also if use T2V generated images for tracking, there will be tracking jumps or failures. We use the sparse image from the two most recent frames t_1 and t_2 as a guide to interpolate and predict the location of the feature point in the current frame. The Semantic Tracking loss is defined as $\mathcal{L}_{ST} = \sum_i \sum_j \left\| \left(T(p_{t_1}) + \frac{(T(p_{t_2}) - T(p_{t_1}))}{(t_2 - t_1)} * (t - t_1) \right) - p_t \right\|_2$ by calculating the displacements of the feature and tracking points.

Furthermore, to enhance the geometric consistency, we utilize SparseCtrl [Guo et al. 2023] and Vectorized Particle-based Score Distillation (VPSD) [Xing et al. 2023] for feedback learning based on the Diffusion model. In comparison to SDS, VPSD addresses issues such as over-smoothing and color oversaturation.

3 RESULTS AND CONCLUSIONS

Our methodology exhibits versatility, as it can be applied to a broad spectrum of complex scenarios. These include various styles, and scenarios with or without backgrounds as shown in Fig. 2.

To demonstrate the effectiveness of our approach more clearly, we select some concise results for comparative analysis, as shown

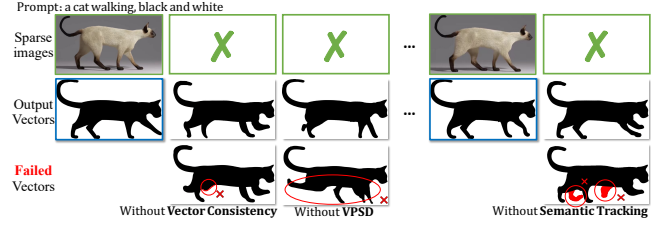


Figure 3: The results and the fail cases.

in Fig. 3. When all three modules are fully utilized, the generated dynamic vectors exhibit general coherence and semantic consistency. To further validate the individual contributions of each module, we conducted a series of ablation studies. Upon the removal of Vector Consistency, we observed that the generated vector exhibited some irregular paths. This anomaly could potentially be attributed to the noise generated during the tracking and diffusion model generation process, a problem effectively addressed by the Vector Consistency. When Semantic Tracking was omitted, the generated vectors displayed some redundant organization, leading to semantic misalignment. Finally, the elimination of VPSD resulted in the generation of some components that were not part of the semantic content due to tracking noise, which could potentially lead to the collapse of the entire model.

Additionally, our method is highly efficient, requiring only 10 seconds of optimization time per frame on A100 80G, significantly reducing the computational time compared to the direct application of the Diffusion Model with SIVE optimization in every frame. This enhances the practicality and broadens the potential range of applications.

In conclusion, our approach can accomplish dynamic vector generation based on text prompts and sparse image control. This methodology ensures coherence and semantic consistency across each vector frame. Moreover, it enables the generation of dynamic vectors for complex scenarios with multiple styles. Moreover, our approach reduces the generation time, thereby facilitating both the generation process and subsequent editing, effectively streamlining the workflow for artists and creators.

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REFERENCES

- Yuwei Guo, Ceyuan Yang, Anyi Rao, Maneesh Agrawala, Dahua Lin, and Bo Dai. 2023. SparseCtrl: Adding Sparse Controls to Text-to-Video Diffusion Models. *CoRR* abs/2311.16933 (2023). <https://doi.org/10.48550/ARXIV.2311.16933> arXiv:2311.16933
- Olga Sorkine and Marc Alexa. 2007. As-rigid-as-possible surface modeling. In *Proceedings of the Fifth Eurographics Symposium on Geometry Processing, Barcelona, Spain, July 4-6, 2007 (ACM International Conference Proceeding Series, Vol. 257)*, Alexander G. Belyaev and Michael Garland (Eds.). Eurographics Association, 109–116. <https://doi.org/10.2312/SGP/SGP07/109-116>
- Ximing Xing, Haitao Zhou, Chuang Wang, Jing Zhang, Dong Xu, and Qian Yu. 2023. SVGDreamer: Text Guided SVG Generation with Diffusion Model. *CoRR* abs/2312.16476 (2023). <https://doi.org/10.48550/ARXIV.2312.16476> arXiv:2312.16476